

Systematic Literature Review

A Systematic Literature Review of Important and Meaningful Differences in the EQ-5D Index and Visual Analog Scale Scores

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ABSTRACT

Objectives: We aimed to provide a comprehensive summary, synthesis, and appraisal of minimally important difference (MID) estimates for EQ-5D instruments.

Methods: We conducted a systematic search using relevant terms related to “minimally/clinically, meaningful/ important difference/change” and “EQ-5D” in 6 major databases, including MEDLINE, Embase, PsycINFO, CINAHL, Scopus, and Cochrane Library (up to January 2023). We included studies that provided at least 1 original MID estimate for the EQ-5D.

Results: A total of 90 studies reporting 840 MID estimates were included. MID estimates for the EQ-5D-3L index score ranged from 0.075 to 0.8 using distribution-based approaches (239 estimates; 20 studies), from 0.003 to 0.72 using anchor-based approaches (189 estimates; 43 studies), and from 0.038 to 0.082 using instrument-defined approaches (4 estimates; 1 study). For the EQ-5D-5L, MID estimates ranged from 0.023 to 0.115 using distribution-based approaches (17 estimates; 12 studies), from 0.01 to 0.41 using anchor-based approaches (97 estimates; 15 studies), and from 0.037 to 0.101 using instrument-defined approaches (62 estimates; 8 studies). For the EQ visual analog scale, MID estimates ranged from 0.96 to 16.6 using distribution-based approaches (87 estimates; 14 studies) and from 0.42 to 51.0 using anchor-based approaches (84 estimates; 24 studies). MID estimates varied by underlying clinical conditions, baseline scores, and direction of change.

Conclusions: A wide range of MID estimates for EQ-5D instruments were identified, highlighting the variability of MID across populations, estimation methods, direction of change, baseline scores, and EQ-5D versions. These factors should be carefully considered when selecting an appropriate MID for interpreting EQ-5D scores.

Keywords: anchor based, distribution based, EQ-5D, minimally important difference, VAS.

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Highlights

- The minimally important difference (MID) has been increasingly used as a method to help interpret changes in patient-reported outcome measure scores in various applications.
- A single, universal MID estimate for the EQ-5D is misleading because of the wide variability influenced by factors such as the version of EQ-5D used, the specific population, estimation methods, and baseline scores.
- Accurate determination of an MID for EQ-5D scores requires careful consideration of these factors to ensure meaningful interpretation of results in any given application.

Introduction

The EQ-5D is one of the most widely used measures to assess health-related quality of life (QOL), offering a standardized measurement of an individual's health status across multiple domains.¹ The EQ-5D has various applications in health-related QOL studies, clinical-effectiveness analyses, and population health surveys and as a patient-reported outcome measure in assessing health system performance.² Given these broad applications beyond economic evaluations, understanding what constitutes a “meaningful” or “important” difference in EQ-5D scores is vital for decision making and policy purposes.³ Solely looking at the EQ-5D scores may not provide a complete understanding of the significance of changes in health status; as such, the minimally important difference (MID), minimal clinically important difference, and clinically important difference emerged as critical and overlapping concepts in this context. MID is the most commonly used term to refer to this concept, and as such it is used throughout the article.

An MID is intended to represent the smallest change in a score deemed meaningful or important to patients, clinicians, and other stakeholders.⁴ It helps to determine whether a change in health status is clinically significant or simply due to measurement error. By identifying an MID, we can better interpret the impact of interventions or treatments on patients' well-being and make informed decisions in healthcare. The estimation of MID involves various approaches, including anchor-based and distribution-based methods.^{5–7} Anchor-based approaches link changes in EQ-5D index or visual analog scale (VAS) scores to an external criterion or anchor, such as patient-reported global assessments or clinical outcomes. Distribution-based approaches analyze the statistical distribution of EQ-5D scores to determine the magnitude of change that exceeds measurement error. In addition, a recent method, the instrument-defined approach, has emerged as a novel way to estimate MID for the EQ-5D index score.^{8,9} This method involves calculating the average absolute difference between the index score of the baseline health state and the index score of all single-level transitions from the baseline health state.

A detailed description of this method has been reported elsewhere.^{8,9}

Several different MID estimates for EQ-5D-3L and EQ-5D-5L index and VAS scores have been reported, exhibiting variations based on scoring algorithms, patient populations, study contexts, and directions of change.^{3,10-15} Using a single value to represent the MID for the EQ-5D index or VAS scores may be inappropriate due to these variations. Furthermore, there is an ongoing debate on the usefulness and legitimacy of using MIDs with preference-based measures of health-related QOL.¹⁶

Through this systematic review, we seek to contribute to ongoing discussions, offering clearer guidance on MID estimation for the EQ-5D. This effort ultimately benefits the multitude of end users relying on EQ-5D for decision making, advancing our understanding of MID estimation and application in health-related QOL assessment. The specific objectives of the review are to identify, summarize, and appraise published studies estimating MIDs for the EQ-5D index and VAS scores, to explore the factors affecting MID estimation, and to identify methodological strengths, limitations, and gaps in the existing literature.

Methods

This literature search and review was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines on systematic reviews and meta-analyses.¹⁷ This review focused on the index scores of the 3-level (EQ-5D-3L) and 5-level (EQ-5D-5L) versions of the EQ-5D and the EuroQol VAS (EQ-VAS).

Literature Search

We conducted a thorough search across 6 major databases: MEDLINE, Embase, PsycINFO, CINAHL, Scopus, and Cochrane Library. The search was completed on January 23, 2023. To ensure a comprehensive search, we collaborated with a skilled librarian in devising the search strategy for each database. The key terms “minimally/clinically, important/meaningful difference/ change,” “EQ-5D,” and their relevant variants and synonyms were used to identify studies reporting estimates of the MID for any EQ-5D instrument. Detailed search terms for each database are available in [Appendix 1](#) (see [Appendix 1](#) in [Supplemental Materials](#) found at <https://doi.org/10.1016/j.jval.2024.11.006>).

Eligibility Criteria

The systematic review incorporated studies that provided at least 1 original estimation of MID for the index or the EQ-VAS score of any version of the EQ-5D. Language restrictions were not imposed during the selection process. However, to ensure precision, we excluded review articles that focused on MID, and any studies, editorials, or commentaries related to the applications of published MIDs of EQ-5D. Given the lack of a quality assessment tool for studies estimating MID, we did not assess the methodological rigor of included studies, and no studies were excluded on this basis.

Review Process

The review process included 2 phases: (1) title and abstract screening and (2) full-text review. Two pairs of researchers experienced in EQ-5D instruments and MID methods (F.A.S., X.L., H.S., and N.S.M.) independently screened the publications and reached a consensus on any disagreements to determine inclusion. Discrepancies were resolved through constructive consensus discussions, and any unresolved discrepancies were

referred to a senior author for final adjudication. The inter-reviewer agreement was evaluated using kappa coefficients to provide a quantitative measure of the consistency and reliability of the review process.

Data Extraction and Management

We used Microsoft Excel (Microsoft Corporation, 2021) to create forms for title and abstract screening and full-text review, and for data extraction and management. We performed pilot testing for all forms for 10% of the included studies at each stage. Subsequently, the remaining 90% of studies that successfully progressed through the title/abstract screening, full-text screening, and data extraction stages were divided into 2 blocks for each of these 3 steps.

Our data extraction process involved a comprehensive examination and recording of information from the included studies. These included general information about the studies (eg, country, data collection year, funding resources), study characteristics (eg, type of population, sample size, study design, sample characteristics), EQ-5D instrument (eg, version, component for which MID was reported, value set used), MID definition adopted, MID estimation method (anchor based, distribution based, instrument defined), anchor type and properties, direction of change (improvement, deterioration, directionless), estimation sample and duration, baseline scores, statistical approach, and distributional properties of the MID estimates and related scores reported in the articles (eg, means, SD, and range).

Data Analysis

Our comprehensive synthesis and summary of the MID estimations encompassed various key aspects. First, we categorized studies based on the estimation methods used, such as anchor-based, distribution-based, and instrument-defined approaches for each component of the EQ-5D instrument: the 3L index score, 5L index score, and the EQ-VAS throughout the article.

For each component, we provided the number of studies and count and range of MID values estimated using each method and presented the median and interquartile range (IQR) of the reported estimates. In the case of studies using the anchor-based approach, a detailed breakdown of MID estimates was presented based on the direction of change, distinguishing between improvements and deteriorations in health status.

In addition, we summarized and presented MID estimates for each EQ-5D score and MID estimation method stratified by clinical condition and baseline scores where such data were available. For anchor-based MID estimations, we evaluated the influence of diverse study designs, including longitudinal and cross-sectional studies, and whether studies were controlled or single-arm trials. In addition, we explored whether the differences used to estimate MID were within or between groups, either at a single time point or over time.

Results

Search Results

Following the PRISMA guidelines, our systematic review began with an initial database search, yielding 4442 records. In addition, 2 records were identified through other sources, bringing the total to 4444 records. After removing duplicates, 1704 unique records remained. Subsequently, all 1704 records were subjected to abstract review, resulting in the exclusion of 1101 records. Following the abstract review, 603 records were assessed for eligibility through full-text review, and 511 records were subsequently

excluded. The reasons for exclusion at this stage included 23 records with unavailable full text, 204 records that were conference presentations, 271 records lacking an original estimation of MID, and 15 records with nonoriginal research study designs. Finally, the systematic review included a total of 90 records that met the inclusion criteria for analysis (Fig. 1). Agreement between reviewers was substantial ($\kappa = 0.79$ for full-text screening).

Summary of Included Studies

Of the included studies, 49 studies reported an MID estimate for the EQ-5D-3L index score (436 estimates), 25 studies for the EQ-5D-5L index score (177 estimates), and 31 studies for the EQ-VAS (173 estimates). In addition, 10 studies reported 53 MID estimates for EQ-5D index score; however, the EQ-5D version (3L or 5L) was not reported and could not be identified through the publication. In terms of MID estimation methods, 43 studies reported using an anchor-based approach to estimate the MID, 9 studies used a distribution-based method, and 6 studies used the instrument-defined approach, whereas the remaining 32 studies used a combination of these methods. The largest proportion of these studies (20 studies; 22.2%) were conducted in the United

States, whereas a smaller number of studies were conducted in other countries (eg, Norway, 8 studies; United Kingdom, 7 studies; Canada, 6 studies; Japan, 6 studies; and The Netherlands, 5 studies), whereas 8 studies involved multiple countries in the sample used. The first study was published in 2005, and the number of publications estimating an MID for the EQ-5D has increased over the years, with a marked increase since 2017 (see Appendix 2 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2024.11.006>). A list of the included studies and a detailed summary of their characteristics are available in Appendices 3 and 4 (see Appendices 3 and 4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2024.11.006>).

MID Estimates for EQ-5D Index and EQ-VAS Scores

A total of 840 MID estimates were reported in the included studies: 436 for the EQ-5D-3L index score, 177 for the EQ-5D-5L index score, and 173 for the EQ-VAS, and an additional 54 estimates for the index score for an unknown version of the EQ-5D (Table 1).

Overall, 49 studies reported 436 MID estimates for the EQ-5D-3L index score with a range from 0.003 to 0.952 (median 0.089;

Figure 1. PRISMA flow diagram.

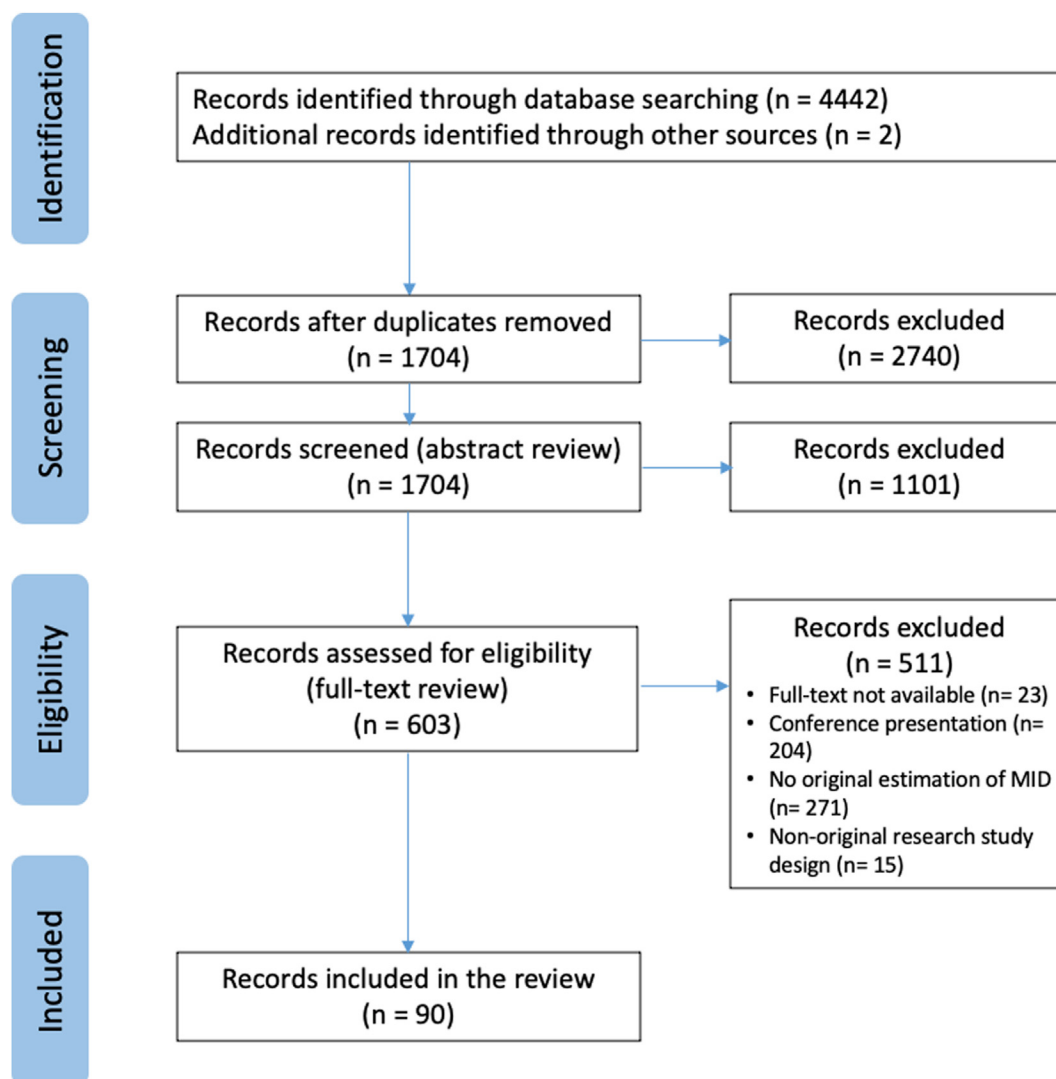


Table 1. Summary of overall results.

Estimation method	EQ-5D-3L index score*	EQ-5D-5L index score [†]	EQ-VAS [‡]
Overall	49 studies/436 estimates Range: 0.003-0.952 (median 0.089; IQR 0.081)	25 studies/177 estimates Range: 0.005-0.410 (median 0.065; IQR 0.057)	31 studies/173 estimates Range: 0.04-51.0 (median 9.0; IQR 5.0)
Distribution based	20 studies/239 estimates Range: 0.02-0.952 (median 0.077; IQR 0.041)	12 studies/17 estimates Range: 0.023-0.115 (median 0.096; IQR 0.048)	14 studies/87 estimates Range: 0.96-16.6 (median 9.0; IQR 4.0)
Anchor based			
Overall	43 studies/189 estimates Range: 0.003-0.720 (median 0.105; IQR 0.121)	15 studies/97 estimates Range: 0.010-0.410 (median 0.060; IQR 0.102)	24 studies/84 estimates Range: 0.42-51.0 (median 8.2; IQR 8.0)
Improvement	30 studies/103 estimates Range: 0.007-0.670 (median 0.130; IQR 0.179)	9 studies/27 estimates Range: 0.017-0.410 (median 0.063; IQR 0.182)	16 studies/65 estimates Range: 0.42-51.0 (median 8.6; IQR 8.5)
Deterioration	13 studies/35 estimates Range: 0.003-0.720 (median 0.090; IQR 0.085)	3 studies/7 estimates Range: 0.010-0.060 (median 0.050; IQR 0.026)	3 studies/8 estimates Range: 0.5-15.5 (median 6.8; IQR 10.5)
Directionless	16 studies/52 estimates Range: 0.025-0.640 (median 0.115; IQR 0.095)	8 studies/63 estimates [§] Range: 0.010-0.300 (median 0.070; IQR 0.090)	12 studies/22 estimates Range: 1.0-41.0 (median 7.5; IQR 8.4)
Instrument defined	1 study/4 estimates Range: 0.038-0.082 [§]	8 studies/62 estimates Range: 0.037-0.101 (median 0.068; IQR 0.030)	NA

IQR indicates interquartile range; MID, minimally important difference; NA, not applicable; ROC, receiver operating characteristic.

*4 additional estimates from 3 studies were reported using a combined distribution and anchor-based approach.

[†]1 additional estimate from 1 study was reported using the anchor-based MID distribution method.

[‡]2 additional estimates from 2 studies were reported; 1 using the anchor-based MID distribution method and 1 using the ROC method.

[§]Median and IQR were not calculated as the number of estimates was <5.

IQR 0.081). For the EQ-5D-5L index score, 25 studies reported 177 MID estimates with a range from 0.005 to 0.41 (median 0.065; IQR 0.057), and 31 studies reported 173 estimates for the EQ-VAS, with a range from 0.04 to 51.0 (median 9.0; IQR 5.0) (Table 1).

Most studies (91.1%) reported the sample size used in estimating the MID for EQ-5D. Sample sizes ranged from 4 to 180 546 (mean 4239; SD 25 761; median 142; IQR 433) in studies using distribution-based methods, from 3 to 180 546 (mean 3073; SD 19 459; median 101; IQR 232) in anchor-based studies, and from 50 to 3351 (mean 1484; SD 841; median 1508; IQR 1011) in instrument-defined studies. More than half of the included studies (54.9%) estimated an MID for patients with chronic conditions (eg, diabetes, asthma, rheumatoid arthritis); 24.2% for patients undergoing surgery, mostly orthopedic in nature; 9.9% for patients with cancer; and 6.6% of studies involves using general population samples. Among all anchor-based MIDs, 59.1% were estimated as within-group differences over time, 32.5% as between-group differences over time, and 8.4% as cross-sectional between-group differences. Global rating of change, (change in) health status, satisfaction, and disease/symptom severity level were used in 60.3%, 19.3%, 13.5%, and 6.9% of the anchor-based MID estimations, respectively. For studies using distribution-based methods, 68.3% reported using some variant of the $1/2$ SD rule,¹⁸ and the remaining used the standard error of measurement for MID estimation.⁴ Detailed summaries of the methods used in the included studies are available in Appendices 5 to 7 (see Appendices 5-7 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2024.11.006>).

Factors Affecting MID Estimates

The MID estimates reported in the included studies varied by the estimation method, direction of change, underlying clinical conditions, intervention, and baseline scores.

MID estimates by estimation method

MID estimates for the EQ-5D-3L index score ranged from 0.02 to 0.952 using distribution-based approaches (20 studies; 239 estimates; median 0.077; IQR 0.041), from 0.003 to 0.72 using anchor-based approaches (43 studies; 189 estimates; median 0.105; IQR 0.121), and from 0.038 to 0.082 using instrument-defined approaches (1 study; 4 estimates) (Table 1; and see Appendix 8 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2024.11.006>). For the EQ-5D-5L, MID estimates ranged from 0.023 to 0.115 using distribution-based approaches (12 studies; 17 estimates; median 0.096; IQR 0.048), from 0.01 to 0.41 using anchor-based approaches (15 studies; 97 estimates; median 0.06; IQR 0.102), and from 0.037 to 0.101 using instrument-defined approaches (8 studies; 62 estimates; median 0.068; IQR 0.030). For the EQ-VAS, MID estimates ranged from 0.96 to 16.6 using distribution-based approaches (14 studies; 87 estimates; median 9.0; IQR 4.0) and from 0.42 to 51.0 using anchor-based approaches (24 studies; 84 estimates; median 8.2; IQR 8.0).

MID estimates by direction of change

For an improvement in EQ-5D-3L index score, 30 studies reported 103 MID estimates using anchor-based approaches with a range from 0.007 to 0.67 (median 0.130; IQR 0.179), whereas 13 studies reported 35 MID estimates for a deterioration with a range from 0.003 to 0.720 (median 0.09; IQR 0.085) (see Appendix 9 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2024.11.006>). For an improvement in EQ-5D-5L index score, 9 studies reported 27 MID estimates using anchor-based approaches with a range from 0.017 to 0.41 (median 0.063; IQR 0.182), whereas 3 studies reported 7 MID estimates for a deterioration with a range from 0.01 to 0.06 (median 0.05; IQR 0.026). For an improvement in EQ-VAS score, 16 studies reported 65 MID

estimates using anchor-based approaches with a range from 0.42 to 51.0 (median 8.6; IQR 8.5), whereas 3 studies reported 8 MID estimates for a deterioration in EQ-VAS with a range from 0.5 to 15.5 (median 6.8; IQR 10.5).

MID estimates by clinical condition

MID estimates for the EQ-5D-3L index score varied considerably across and within clinical conditions (see [Appendix 10 in Supplemental Materials](#) found at <https://doi.org/10.1016/j.jval.2024.11.006>) ranging from a median of 0.07 ($n = 165$ estimates; IQR 0.04; range 0.02–0.18) in cancer, 0.082 ($n = 54$ estimates; IQR 0.036; range 0.012–0.34) in neurological disorders, and 0.14 ($n = 145$ estimates; IQR 0.179; range 0.007–0.952) in musculoskeletal disorders. Similarly, MID estimates for the EQ-5D-5L index score ranged from a median of 0.101 ($n = 66$ estimates; IQR 0.15; range 0.01–0.41) in musculoskeletal disorders, 0.046 ($n = 19$ estimates; IQR 0.039; range 0.005–0.109) in respiratory diseases, and 0.049 ($n = 29$ estimates; IQR 0.03; range 0.021–0.089) in endocrinological disorders. Similar variability was observed in the MID estimates for the EQ-VAS across clinical conditions, whereas the median estimate was 9.0 ($n = 70$ estimates; IQR 4.0; range 2.0–14) in cancer and 10.3 ($n = 52$ estimates; IQR 14.2; range 0.04–51) in musculoskeletal disorders.

MID estimates by intervention

For the EQ-5D-3L index score, MID estimates in cancer studies involving surgical interventions ($n = 21$ estimates) ranged from 0.05 to 0.18 (median 0.14; IQR 0.04), whereas in those with no surgical interventions ($n = 142$ estimates), MID estimates ranged from 0.02 to 0.16 (median 0.07; IQR 0.04). In studies that estimated MIDs in orthopedic surgery ($n = 98$), estimates ranged from 0.01 to 0.952 (median 0.186; IQR 0.2), whereas estimates in nonsurgical studies ($n = 43$) ranged from 0.007 to 0.72 (median 0.082; IQR 0.099). For the EQ-5D-5L index score, MID estimates in studies involving orthopedic surgery ($n = 22$ estimates) ranged from 0.027 to 0.41 (median 0.175; IQR 0.146), whereas in those with no surgical interventions ($n = 39$ estimates), MID estimates ranged from 0.01 to 0.22 (median 0.07; IQR 0.09). For the EQ-VAS, MID estimates in studies involving orthopedic surgery ($n = 43$ estimates) ranged from 0.04 to 51.0 (median 11.0; IQR 14.7), whereas in those with no surgical interventions ($n = 8$ estimates), MID estimates ranged from 0.74 to 23.1 (median 9.2; IQR 13.2).

MID estimates by baseline score

MID estimates for the EQ-5D-3L and EQ-5D-5L index scores were correlated with their respective baseline scores (3L: $r^2 = 0.2$; 5L: $r^2 = 0.5$) (see [Appendix 11 in Supplemental Materials](#) found at <https://doi.org/10.1016/j.jval.2024.11.006>). As baseline scores increased, the MID estimate decreased. This trend was more evident in the MID estimates for the 5L than the 3L index score. MID estimates for the EQ-VAS did not seem to be affected by the baseline scores.

Discussion

Through this systematic review, we aimed to contribute to the understanding of MID estimates for EQ-5D scores used at the group level, offering guidance on their use and implications for decision making in health-related QOL assessments in various contexts. The comprehensive analysis of 90 studies and 840 MID estimates allowed for a nuanced exploration of MID across different versions of the EQ-5D and various estimation methods. Our review identified a wide range of MID estimates for EQ-5D instruments in the literature, demonstrating that estimates vary

among different approaches, direction of change, anchors, clinical conditions, and baseline health status. This finding reinforces the notion that a one-size-fits-all approach to MID is inadequate, and careful consideration of context-specific factors is crucial. Moreover, the findings of this review suggest that the term “minimal” difference may be misleading and inappropriate for the applications of this parameter and that “meaningful” or “clinically important” difference or change could more accurately describe this concept. Future research should reconsider the appropriateness of these terms.

Three main MID estimation methods—anchor based, distribution based, and instrument defined—were used, with some studies combining methods. Anchor-based approaches, linking EQ-5D changes to external anchors, dominated, followed by distribution-based methods and the emerging instrument-defined approach. Notably, anchor-based approaches reported wider ranges than the other 2 methods, whereas the instrument-defined approach yielded the narrowest range of values for the EQ-5D-3L and EQ-5D-5L index scores. The prevalence of anchor-based methods aligns with the established practice in the literature and suggests a reliance on external benchmarks for MID determination.⁶ More than half of the MID values identified in this review were estimated using the anchor-based approach, which significantly contributes to the wide variability observed in these estimations. This variability arises from the diverse range of anchors used, the various statistical methods used, and the differing timeframes used in assessing change.^{19,20}

The findings of our systematic review align with the existing literature, which has reported variations in MID estimates for EQ-5D-3L and EQ-5D-5L.^{3,9,10,12–15,21–23} The variability observed in MID estimates can be attributed to diverse scoring algorithms, patient populations, and study contexts, consistent with previous research.^{3,9,10,19,20} The exploration of MID estimates by clinical condition and by improvements and deteriorations in health status revealed substantial variability, emphasizing the need for context-specific MID values. Notably, MID estimates for improvement were generally larger than those for deterioration in EQ-5D-3L and EQ-5D-5L index scores and in EQ-VAS scores. Furthermore, MID estimates for the EQ-5D-3L and EQ-5D-5L index scores or EQ-VAS scores generated in studies involving a surgical intervention were larger than those estimated in studies that did not involve any intervention. This observation was consistent irrespective of the nature of the intervention.

The multitude of MID estimates reported across studies underscores the complexity of defining a universal MID for the EQ-5D. The substantial impact of the estimation method, direction of change, clinical condition, intervention, and baseline scores on MID values highlights the necessity of considering these factors when selecting and using an MID for interpreting EQ-5D scores. For instance, the correlation between baseline scores and MID estimates, particularly evident in EQ-5D-5L, suggests that patients with higher baseline scores may perceive smaller changes as meaningful. This has important implications for interventions targeting populations with varying baseline health statuses. Users of the EQ-5D instrument aiming to use an MID for interpreting differences or changes in scores or the impact of interventions must carefully select the MID, taking into account the aforementioned factors to identify the most appropriate MID for their target population. In addition, given the inherent complexity, users should consider conducting a sensitivity analysis using various MID values. In cases where variability is substantial, it may be advisable for users to estimate an MID value based on their own data, provided that the necessary data elements and design are available.

Although various methods for analyzing and interpreting EQ-5D dimensions and summary scores exist, MIDs are increasingly

gaining a role in interpreting EQ-5D data in applications, such as assessing treatment effectiveness, evaluating programs, and comparing healthcare provider performance in routine outcome measurement initiatives within many health systems. As EQ-5D data collection becomes routine in healthcare systems, relying solely on statistical significance risks overlooking differences of small magnitude. Establishing standardized guidelines for the estimation of MIDs in EQ-5D instruments and other patient-reported outcome measures is essential for enhancing the interpretation of patient-reported outcome measurements and upholding methodological rigor in MID estimation studies. Such guidelines are crucial to provide a consistent and reliable framework, promoting accuracy and comparability across diverse research contexts and ultimately advancing the meaningful interpretation of patient-reported outcomes.

Methodological Strengths and Limitations

The review followed PRISMA guidelines, ensuring a systematic and transparent approach. The collaboration with a skilled librarian in devising search strategies enhanced the comprehensiveness of the literature search ensuring proper capture of the existing literature. However, the lack of a quality assessment tool for MID studies is a notable limitation, which led to including all types of studies irrespective of their methodological quality; this may have affected the quality of the pooled estimates reported in this review. Future research might benefit from the development of a standardized tool for assessing the methodological rigor of studies estimating MIDs. In addition, the exclusion of non-English studies may introduce language bias.

Conclusion

This systematic review provides a comprehensive overview of MID estimates for EQ-5D, highlighting the complexities and nuances involved in determining a universally applicable MID. The observed variations in MID values across different versions of EQ-5D, estimation methods, and clinical conditions underscore the need for a context-specific approach. Moving forward, researchers and decision makers should consider the diverse factors influencing MID estimates when interpreting EQ-5D scores and making informed decisions in healthcare. As the field continues to evolve, ongoing dialogue and methodological refinements will be essential for advancing our understanding of the MID in the context of this widely used instrument.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

Supplemental Material

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.jval.2024.11.006>.

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